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H8SCOOP

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by Henry Fale

BITS & PIECES

HAPPY ANNIVERSARY

This is the 1 year anniversary of H8SCOOP. The newsletter has been quite a success, and continues to grow in content and circulation. My first 8 page issue occupied about 90 sectors on disk storage, and my last one occupied about 160 sectors. You will continue to get 12 issues a year. When there is more to print, it will just be a larger issue, at no extra cost to you. H8SCOOP has been rated number one and I intend to keep it that way. The addition of the Diablo 630 has allowed me to get more information per issue, and get it printed faster with better quality.

I plan to add CP/M WORDSTAR or MAGIC WAND in the near future, which will be still another improvement in the quality of H8SCOOP. You can also look forward to more software marketing--some really good stuff coming in the future.

I wish to take this space to thank all those who have contributed to H8SCOOP in the past year; articles, ideas, and software. Without you this newsletter would not be what it has been. I look forward to your continued support. I plan in the future to be able to start compensating authors for their work, and also for the software contributed, more than I have been able to do in the past. I also will be compensating those who have already contributed software, depending on the volume of disks sold.

Z89 FOR H8SCOOP?

It's true, this edition is being prepared on a P4 green screen loaded Z89 computer! No I haven't been in the sun too long, I got a good deal on this unit and didn't even know it was green screen till I got it home and turned it on. My heart skipped a beat! The excitement of it all. I knew the green screen was coming, but I wasn't expecting it yet.

I'm going to sell and totally support business systems consisting of the Diablo 630, Z89, and 8" drive systems. No, I'm not selling out the H8! I have two of them and are adding on to them, but for a business system where the guy doesn't want the extras, and for a system I can take with me to work on H8SCOOP the Z89 is just right.

MY VIEW ON AUTHOR ETHICS

There have been readers and authors who have written to me in the past on the policy of an author submitting the same material for publication to various newsletters. To me this is a definite NO! I have had negative comments in the past that

an identical article was printed in H8SCOOP, BUSS, and REMark. The publisher has no control over this, as he does not know what the rest have received and are printing, until the newsletter or journal is out.

I have already been in the situation where H8SCOOP was all layed out and ready to go to press. I received BUSS, and found two identical articles contained within, so I had the painstaking task of deleting the material from H8SCOOP, and writing other articles that would fill about the same space as the deleted articles. This was a many hour task. Most of my readers receive at least one, and usually two or more of the other newsletters, and it's wasted money to open the current issue of a newsletter only to find you've already read the material.

The solution is obvious. Authors should only submit their material to one source at a time. If you want to rotate, fine. If you have a favorite, fine. This should prevent any problems and hard feelings in the future, especially now since H8SCOOP is copyrighted! My policy stands as stated before. Those who heavily contribute to H8SCOOP will receive a kickback of some sort, either by subscription renewal (as has been past policy), or for major contributions, there will be a cash award. After all, it's work for you and you should receive something.

Anybody wishing to submit an article or any information, if you want to know if it will be used, I will immediately acknowledge by return mail, if I will be printing it, and when. This way you know where you stand and if I won't be using something, or may be delayed too long, you may choose to submit it to someone else. Fair enough?

H8SCOOP PHONE HOURS

I would like to call your attention to my Phone Hours as printed on the last sheet of each H8SCOOP. I ask that you note these hours when calling, as I'll try to be available at these hours as much as possible. Any other hours, you call at your own risk of my not being available and your wasted call. The only thing I will warn you, I WILL NOT accept any calls between the hours of Sundown Friday night, and Sundown Saturday night, as this is the Bible Sabbath, the Lord's Day, which I observe--absolutely NO BUSINESS during those hours (Ref: Exodus Ch 20 Vs 8-11).

NO MORE PORTZABEE

OK, many of you have written in the past wanting to know what PORTZABEE was all about. I'll tell you. It really has no significance. When I was in college, I built a small training digital computer. I was into "MAD" magazines at the time, and often

there would be a sign saying "This way to Portzabee". I thought that was cute, and it stuck with me. I named that first computer of mine, the PORTZABEE I, and later developed another, which I named the PORTZABEE II. When I started my computer support business about 4 years ago, that name still stuck with me, so I used it for my company.

Now I'm putting PORTZABEE to bed and have formed a corporation, under which H8SCOOP is a subsidiary, named QUICKDATA COMPUTER SERVICES, INC. This is the name you'll see from now on (at least once I get the PORTZABEE letterhead used up!). That is the name of my Z89 business system support business.

ON GLARE SCREEN FOR H19/H89

I purchased the HCA-4 Fabric Anti-Glare CRT filter from Heath (\$19.95) and would strongly recommend it to anyone with the H19 or H89 that wants a better look from the screen. It improves the contrast and readability drastically, and is not hard to install. The CRT does not have to be removed, only loosened. The quality of the screen starts to get close to the Hazeltine, which is an excellent screen.

H8 MUSIC SYNTHESIZER BOARD

I also purchased the HA-8-2 stereo music synthesizer board from Heath (\$159) after hearing the thing work. It's really fantastic, with excellent 4 voiced stereo music quality. It's one good way to show off the H8. If you know anything about reading music, it's a breeze to program your own programs with the included utilities. It comes with a disk with music already on, plus quite a bit of documentation. The only disappointment was the schematic was not included, which I feel is essential, especially since they claim you can use it for A/D conversion also.

ON TED BENGLIN'S TALKING H8

I had one of Ted's units wired to a parallel port of my computer for a few weeks, and it was amazing! I had no intention of purchasing one, but after hearing this thing, and seeing the ease of programming it, I called Ted right up and ordered one.

One great thing about it, though not too apparent at this point yet, is the software support being planned for it. I won't say much about it now, but there are gobs of software coming to simplify programming it and making full use of it's capabilities. There will be a driver available to interface it with Don Long's PC-12 clock/parallel board, which will make it more popular, since that is THE board to have. A talking clock should be forthcoming.

Ted sent me some update literature on the talking H8, MI-8. Because of the excellent response, he just purchased a large quantity of VSL Modules and will be able to pass along a \$75 reduction in all prices with the Module, kit and assembled! The prices are now as follows: Completely built unit with housing is \$575; Completely assembled V-8 board with power supply and VSL Module for \$499.95, ready to plug into H8-2 parallel board; Complete kit \$424.95; Basic kit \$149.95; PC Board and Doc

\$67.95; Documentation only, \$49.95; And Demo Voice Cassette for \$5. All orders except the Demo Cassette come complete with documentation and include a demo/diagnostic program on disk.

Because of the on-hand supply of VSL Modules, there is only a 2 week delivery on assembled units and kits. The included VBDVD device driver now contains SET options for changing PORT number using simple SET command. For more info or to order, contact Ted Benglen, 822 E Country Rd. 30, Ft. Collins, CO 80525. (303) 669-4116.

H8 STORAGE ROOM TIP

Tom Rule writes in "After installing a fan in the H8 computer, it ran fine for several months without any heat problem. Hence, not opening the H8 computer for a long period of time invites me to clutter the top with cassettes and notes.

"I picked up 1/8" thick plywood, cut it to the same size as the top of the H8, glued one quarter round molding around the edge, making the plywood a tray. Cut 3/4" dowel into four pieces for legs, 3/4" long each, and glued each of them to the corners where they fit inside the top edge of H8 sides.

"Now the tray is on top with air space of 3/4" for fan to pull air in and the tray is cluttered now! When I need to open the H8, I take the tray with it's contents away!"

FORTH INTEREST GROUP INFO

The FORTH interest group has over 2500 members worldwide. Fifteen chapters are meeting or in the process of forming. In keeping with the spirit of the FORTH language, several groups meet on the fourth Saturday each month.

All members receive the bimonthly publication, FORTH DIMENSIONS. The list of publications from FIG (FORTH Interest Group) has expanded greatly over the past year. There are now seven different source listings for specific CPU's with several more being developed. A number of user's manuals can be obtained as well as FIG T-shirts and posters.

1981 looks like the year of FORTH. A number of manuals and books are in the works. More and more vendors are selling FORTH implementations and programs. The utilization of FORTH is spreading from micros to robotics, business applications, product development and many control applications. FORTH DIMENSIONS will keep members abreast of these activities. Memberships are \$12/year (\$24 foreign) and include six issues of FORTH DIMENSIONS. For a list of publications and to become a member, contact FORTH Interest Group, POB 1105, San Carlos, CA 94070. (415) 962-8653.

Yes Virginia, there is a FORTH for the H8 and H89 under HDOS. See NEW STUFF.

ON FLOPPY DISK SERVICES DRIVES

Jeff Dovel writes in "I took your suggestion and called Floppy Disk Services Inc (See H8SCOOP November 1980). They were a week behind on orders, but did have a FDD-100 5 for \$235." Jeff ordered it and found it was capable of writing on both sides

of the disk without having to cut out any holes in the diskette jacket, just turn the diskette over.

It's true! I read their latest advertisement, and saw the "flippy" which will allow the disk to be turned over on the other side.

ON ANDEX COMPILER

James Czebiniak writes in on the Andex Compiler "I think that the advertising for the Andex Basic Compiler should be clarified. My definition of an MBASIC compiler is one that will take a running MBASIC program and create assembler or object code. This is not the case with the ANDEX COMPILER. It is an MBASIC program that converts a very restricted set of non restrictions which make it a little difficult to use and only supports integer arithmetic. This compiler is a very good super-macro assembler and is supported well by Mr. Dessler. I found a bug and they sent me a fix in three days. My only problem is that the advertising was not explicit enough (even the blurb that I sent for) and after spending \$50 I found it could not do what I bought it for."

ON HA8-8 H8 0 ORG BOARD MOD

I was told there are some problems in the Extended configuration board which allows the H8 to run with 64K, and required for use with the new 0 org CP/M and the dual 8" drives. It seems this board will work for 48K or less, but not 64K. I have this board but have not installed it and tested yet. Any more comments on this, I'd appreciate it. Word has it that a new board is being worked on to take care of this problem. Does anybody have a shipping date for that? What about the rest of us that have the one that doesn't work? Rick Martinek from Milwaukee Heath Store says the one he sold me (HA8-8) will work. He should know, he eats 0-org CP/M for snacks, and has among other computers in his house, a giant IBM.

SHURGAT HARD DRIVE KIT

Shugart Associates, 475 Oakmead Parkway, Sunnyvale, CA 94086 (800) 824-7888 and in California, (800) 852-7777; ask for operator 112. They have available a SA1000 Winchester Success Kit for evaluation purposes for \$1950, which is about \$1000 off normal price. It includes a 5.3 Meg SA1002; a SA1400 intelligent fixed/floppy controller, interface cables and support doc.

It looks very good. Anyone involved in designing good things for the H8 and H89 that understands assembly language software at disk programming level should take advantage of this limited offer. It looks like for about \$2000 hard drive for the H8 or H89 is possible with this setup. Contact Shugart for more info.

PROBLEM WITH PAGE EDITOR

It was reported that PAGE, the HUG editor may have problems with the ESCape sequences when operated at less than 9600 baud. To remedy these problems, simply change your baud to 9600.

ON H19/H89 SCREEN GRAPHICS PROBLEMS

There appears to be a problem with the screen display of the graphics terminals when working with ESCape sequences at times. Characters may be inserted at various spots instead of graphics, and lines and columns may not line up. This problem has been traced to two faults. First, for those who have the early terminals capable of operating at 19200 baud, this speed is known to screw up the escape sequences, and should not be used when working with any graphics. That is why the new terminals support a maximum of 9600. Any problems, start by switching your BAUD to 9600 or less. 9600 seems to be the standard used most.

The second problem deals with wrap around mode. If you have the terminal width set at 80 or less, it may pose a problem. To get around this, use the SET TT: WIDTH 255 command to increase it, with 255 you should be safe.

ON H9 KEYBOARD FROM GRI

Despite all promises and talk, I understand the keyboard mod for the H9 is still not available from GRI (see March '81 H8SCOOP). Russel Miller tried to get one, and was told at the end of February, and then the end of March. They told me it would be out last year already! I think we better BEWARE of GRI and not get our hopes up for that H9 mod. I venture to guess they are having major communication or administration problems, and should be avoided in either case.

NATIONAL TECHNICAL SCHOOLS H88 DELAY

Due to FCC regulations on RFI, Kit production of H8's, H88's and H89's have been halted to allow for redesign and re-tooling. This has caused a delay in the National Technical Schools computer program where the H88 is being offered as part of the course. Shipment is expected no earlier than April.

ON MICRONET

Individuals not already on MICRONET may have a hard time getting on it, as it is becoming increasingly difficult to get in unless you are with Radio Shack. Possibly if there is enough interest, and the difficulty continues, you could get together with your local HUG'S and try to put the pressure on.

AUTOSCRIBE OWNER PO'ED

Jan Johnsen called to tell me, that although the new release of AUTOSCRIBE, Heath/Zenith's word processor is being released and shipped to new customers, he, as a registered owner of a previous issue, has not received the promised updates, nor heard about it yet. After shelling out \$400, you'd think they would support what's already out there rather than rushing to sell more.

NEW STUFF

BUSINESS SOFTWARE AVAILABLE

S + M systems Inc, POB 1225, 2 Washington St, Haverhill, MA 01830 (617) 373-1599 has available a line of business software for the Z89. Since the software is written in MBASIC 5.1 or 0 Based CP/M 2.x, I see no reason why it won't run on the H8 as well.

They unconditionally guarantee that the system they sell is BUG FREE. Included in their software are WORD PROCESSING, ACCOUNTS PAYABLE, GENERAL LEDGER, INVENTORY CONTROL, CLIENT BILLING, ACCOUNTS RECEIVABLE, and PAYROLL. Typical prices are \$125 for programs, which can be used with dual 5 1/4" drives and 8" drives. For more info, contact the above.

PASCAL REFERENCE CARD AVAILABLE

From Modular Software, POB 12883, San Antonio, TX 78212 comes a general PASCAL reference card. It is packed full of data, including Reserved and predefined words, Operators, P-Machine OP Codes, Transcendental Functions, ASCII Character set, Functions and Procedures and more. For those into PASCAL, this handy reference will be useful for a fast glance to find what you want. The cost of this card is \$2.00.

GOOD DISK/MEMORY CHIP PRICES

Earl Albin tells me Memorex 10 sector disks are available for \$22/box from Mark Gorden Computers, 1-800-343-5206.

WW Component Supply, INC, 1771 Junction Ave, San Jose, CA 95112 (408) 295-7171 has 4116 equivalent memory chips, MITSUBISHI 4116-M58759-20 which is 150ns chips, 8 for \$28. They also have the National MM5290N-3 which is the 200ns chip, 8 for \$24.

ON METAL ENGRAVING PLATES

From AURORA ENTERPRISES, 8703 Edgewater, Warren, MI 48093, (313) 939-1513, the same ones who introduced the metal MODES OF OPERATION plate for the H19, comes more metal engraved items. Modern technology has given the time honored art of engraving a new outlook and new possibilities by introducing electronics into the procedure. You can have almost any written word or picture reproduced onto metal.

Almost anything you may have, business cards, door signs, desk nameplates, paperweights, photographs, diplomas etc, can be engraved into a metal plate.

Electronic Metal engraving is a process whereby the item to be copied is scanned with a beam of light. The beam sends an impulse to a carbide tip located above a special metal plate which is then engraved with exactly the image seen by the beam of light on your original copy. The cost for this service is 30 cents per square inch of metal, with maximum size of 12" X 18" in various colors. For more info, contact AURORA.

SOURCE FOR COMPUTER "OFFICE" SUPPLIES

For items such as Selectric type balls and ribbons, Diablo print wheels and ribbons, mailing labels, burst binders, and much more, THE RELIABLE CORP, 1001 West Van Buren St., Chicago, IL 60607 has some good prices. Call 1-800-621-4344 out of state, or (312) 666-1800 Illinois and ask for the Computer Supply Sale Catalog. Tell them you heard about it from QUIKDATA.

PROGRESS REPORT FROM TRIONYX

Bill Perry of TRIONYX writes "A major obstacle to the addition of new high- technology products in the H8 computer is the H-50 buss implementation. The H8 motherboard is poorly designed and does not function reliably. Board position sensitivity is easy to demonstrate." They are offering a new motherboard for the H8 which is being designed by an electrical engineer and comprise a proper set of computer-bus signal transmission lines. This new motherboard the T-H90 will sell for \$39 without parts. Gold plated connectors and front-end termination resistors will be extra. This will be a 90 pin bus with 80 signal lines on one side of the board and a ground plane on the other. It should be available around the time you read this.

The Ground plane is essential for reliable 4 MHz operation. Heath is not going to offer 4MHz operation on their new Z80 CPU board apparently because they feel their bus cannot support it.

This new motherboard will be completely compatible with all boards presently used in the H8. Trionyx plans to use the extra signal lines on the new bus for some of their new H8 products forthcoming.

A rear-end bus termination card will also be made available to not only terminate the bus, but contain the drivers, receivers and connector necessary for an expansion box cable.

Modification kits will be offered to convert each of the standard Heath H8 cards to their bus. This conversion is optional, but the addition of extra ground connections to the bus on each card should enhance the reliability of the computer. The five-pin connector on the Heath CPU board is very inconvenient and can be eliminated using the extra connections on the Trionyx new motherboard. They are presently working on memory management for the H8 to extend its addressing capability beyond 64K. Some of the extra lines on the new bus can be used for memory bank select. They are also working on a 16 bit CPU board, with 4MHz, which will quadruple the present speed of the H8.

The substitution of the new bus motherboard in the H8 will also present a convenient opportunity to change from the unreliable tin plated to gold plated connector pins; an absolute requirement for reliable system operation.

TRIONYX intends to be 100% Heath hardware and software compatible at all times. All of the new products offered will be "upward compatible".

Bill tells me their new Z80 CPU board has slipped schedule and probably will not be ready until the end of April, but will have a number of special

features and worth the wait.

MISCELLANEOUS

I got some info back from JWS Engineering (see last month) on the Z80 board for the Heath H8. The blurb says "JWS Engineering is announcing a Z80 board, compatible with the H8 buss. The board contains a Z80 CPU, 8K eeprom-2716(4), and circuits required for H8 buss interface. Power on jump to any 1K boundary. The bare board is \$37.95, and the kit price is not listed. That's all it says. JWS Engineering, Box 67, Lebanon, NJ 08833.

Don Long plans in late April to have Kits available for his clock/12 channel parallel board. It's important to note, Don plans full support for this board, with many applications under way so as not to leave you high and dry. He also still has available a schematic interface for the SCI BSR controller for the H8 computer. Any questions or to place orders, contact Don, 9319 Willowview, Houston, TX 77080.

Questions have been coming in on Jerry Phelps Space INVADERS program for use with Greg Saville's Sound board (for H8 only), as to the type of sounds produced. Some of the sounds produced are: footsteps, explosions, laser beams, death rays, spacecraft, bombay doors, and more. The program is sent on a disk and the price of \$17.50 includes everything; disk, postage, diagrams, and instructions.

Steve Donahue writes in really praising this version of INVADERS. "I received the disk from Jerry promptly and was pleased at the design of the program which was able to run using MBASIC 4.82 and HDOS 1.6, or with 1 minor change 2.0. I can appreciate the time Jerry has spent on this program since so far about all the effects I have used are from the demonstration program which Greg Saville supplied with his board. The potential is there but a lot of work is involved in matching sound and display, which Jerry has effectively done."

Word has it that Heath is working on hard disk for the H8 and hope to offer a hard disk/8" floppy combo in an H-47 cabinet sometime this year.

SSGT James Romoser sent me some information from DISKS, ETC., POB 327, Center Valley, PA 18034 (215) 439-1717. They have a very complete line of computer related products. Printer fan-fold paper, lables, binder for fan-fold paper, disks, disk mailers, disk storage cases, and more. For \$34.30 they have a locking diskette storage case with key lock, which will hold up to 100 mini-floppies. Write to them for a list of their products.

JIM TEIXEIRA, 62 Churchill St., Sudbury, MA 01776 has lot of good software available on disk. He now has his famous SYSMOD out for HDOS ver 2.0 for \$17.95--it's called SYSMOD2. Of course the SYSMOD for 1.6 is still available and is priced at \$15. These modify the system to provide more and better features and make it easier to type commands (See February 1981 H8SCOOP Pg 2).

Besides these, he has BASEDIT, a program to give you a built-in editor for HDOS BASIC, similar to the MBASIC type of operation. This allows you to

change part or parts of a line without having to always retype the entire line. It sells for \$15.

Jim's DIAS is a fast disassembler written in TINY PASCAL, which generates source code suitable as input to the Heath Assembler. The Tiny PASCAL source code is included for this disassembler, along with TINY PASCAL and its instructions. This sells for \$25.

EDGE is an improved Text Editor from Jim for \$25, available either for the H9 or H19 systems. This is a HDOS text Editor/Generator which is a vastly improved version of the public domain editor describer in REMark 6.

And Last, something many have been looking for, HFORTH. This program is based on the fig-FORTH ver 1.1 developed by the FORTH INTEREST GROUP. It has been modified to run under HDOS and makes extensive use of the HDOS file structure including allowing named files for both program and data disk I/O. It also includes a built in editor for generating and editing HFORTH programs. Looks like a good starter package, or for those wanting to know about FORTH. Incidentally, most of these are available from Heath stores. The price of \$29.95 include postage, but overseas add \$2.00.

J.J. Thompson, 281 Warren Ave, Kenmore NY 14217 is offering new versions of the HDOS to CP/M and CP/M to HDOS file conversions HOSCOPY and CPMCOPY. These are menu driven and allow you to specify the drive used for source and destination disks, to change disks without leaving the program, to list the directory of either or both disks, and to specify file name for destination. The HOSCOPY disk contains versions for 0 ORG CP/M 2.2 and 4200 ORG CP/M 1.43. CPMCOPY will run either HDOS 1.6 OR 2.0 and read both 0-ORG and modified format CP/M disks.

These new versions will cost \$20 plus \$1 shipping in US and Canada and \$3 shipping foreign. Current owners of this software can update by returning the original distribution +\$10 for each or \$15 for both, if returned together, plus postage.

From William Hall, 5621 Maple Heights Ct, Pittsburgh, PA 15232 comes AUTOSEND, a MORSE CODE practice program for the H8. This program will give you random characters (depending on the categories you select) at selectable speeds. You copy the characters as you hear them, and check with the screen display on your accuracy. It runs under MBASIC. Selling price is \$15 which includes disk, handling and postage. To order, or for more info, contact William Hall.

From Dr. Jay Gold, POB 2024, Des Moines, IA 50310 comes an extensive HOME FINANCE SYSTEM to run on H8 and H89. Comes complete with 109 page manual. It appears to be quite extensive, and has been over 2 years in the making. I'll let the Doc tell you about it.

"This system keeps track of all your banking activities. All of your checks and checking deposits, as well as your savings activity (for up to two banks) may be recorded on a single floppy disk. The system also allows you to define recurring payment accounts (like credit cards, utilities or department stores) that send you

regular bills. When you pay bills, you can enter the billing data once, and the data will be stored in your check register and in a detailed record of your transactions to that account. When you are done, your checks will be printed out on your printer for you if you wish. Just sign them and pop them in the envelopes!

"All of your checks have a code that you define. You may then examine your monthly expenses according to this coding system. I have provided extensive display and printout functions to help evaluate your financial activity.

"I have designed this package particularly for the user who wants the computer to preform a task, but who may not be a computer expert. The 109 page, loose- leaf manual that accompanies the program disk is written with a minimum of computer jargon. It explains first in general terms, and then in considerable detail, all the functions of the system. Extensive error handling is incorporated in the programs so that the novice should not be concerned that slips of the fingers would bring catastrophe.

"This Home Finance System requires the H8/H19 or H89 computer with two 5 1/4" drives, 48K memory and either the H-14 or H-44 (Diablo) printer. The system consists of a 109 page users manual in a loos-leaf binder, and four disks. Over thirty BASIC and assembly programs reside on the first three disks. The fourth is a Sample Data Disk that allows the user to experiment with the system before committing his or her own data to it. I am offering this System for \$85.00 plus shipping. The manual may be purchased separately for \$17 plus shipping, and this amount will be credited towards purchase of the complete system."

OFFERS

I make all software available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy disks. You take your chances. At my prices, you really can't go wrong. I try to get software out as fast as possible so you can get the newest stuff fast, and as thus, don't get a chance to carefully screen and test everything. To do so, would delay all software at least 3 to 4 months. I do the best I can and sell without assuming responsibility on the accuracy or runability of a program or disk. I don't think I've let anyone down in the past and will continue to do my best not to.

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested readers.

TINY PASCAL	\$10
SEPT DISK--FILE DUMPS & UTILITY	\$ 7
NPS CP/M MICRO COBOL SET(2)	\$20
NPS CP/M MICRO COBOL SAMPLE TEST PROGRAMS	\$ 5
RATFOR TRANSLATOR (MUST HAVE FORTRAN)	\$ 7
NOV. DISK--PICTURES, LABELS, WORD PROCESSOR	\$15
QUICKMAIL MAILING LIST PROGRAM	\$15
C SOFTWARE STUFF DISK	\$10
DEC. DISK--PICTURES & UTILITIES	\$10
H19 GRAPHICS PROGRAMS DISK (1/81)	\$10

2/81--PLANT, SWARMS, STARWARS, UTILITIES	\$10
3/81--POTPOURRI SET(2), misc cassette conv.	\$15
3/81--ADVENTURE SET(2), PIRATES, LOST DUTCH.	\$15
DOUBLE LA-36 SPEED TO 600 BAUD ARTICLE	\$ 6

Foreign orders add \$2/disk for postage and handling.

Mike Miller writes in "A word of caution for your readers using UNLOCK from your FEB. disc. The program apparently corrupts a disk with HDOS 1.5 system files on it, making the files inaccessible." I also found out it won't work on 2.0 --Henry.

Richard King wrote me that the RATFOR disk I sent him would not boot. He discovered there was a bad SYSCMD.SYS. When he substituted this, everything worked fine. I thought I'd pass that along in case anyone else was having that problem.

This month I offer the APRIL '81 disk, THINGS for \$10. Things is mainly for use with H19 or H89 and MBASIC. It does make use of lot of graphics and these will not run on any other screen. There are other files that you may be interested in. DB.DVD is a LP.DVD driver which allows the Diabole 630 to print bi-directionally by modifying the LP from 32 character burst to 600 character burst (per Walt Bilofsky's idea). This gives the Diablo internal character buffer more characters so the micro processor can "look ahead" enough to do bi-directional printing. Another .ABS file is ERREP which when called from HDOS 2.0 gives the "missing" STATUS report on read/write's, and Hard and Soft errors (See Technical Forum). Another .ABS is BOGGLE, the word game (similar in function to program published in June '80 H8SCOOP). Still another is DIRSORT.ABS which places your disk directory in alphabetical order.

In Graphics there are two more Stupid Fascination Programs, A few excellent graphic animation programs (TOWER and DAISEY). There is a TRIANGLE.BAS in BH EXT BASIC for calculating Right Triangles. This draws a Triangle on the screen and fills in the parameters (length and degrees) as you enter the known values. It then finds the unknowns. A Non-graphic is BIORITHM which is quite complete and short, runing under MBASIC. And MORE!

WHO'S WHO

SCALL is a good local HUG newsletter dealing with Heath computers and the news of the Houston Texas HUG (HUG-H). Membership to HUG-H is open to any person interested in Heath brand computers. Dues are \$12/year, which include a subscription to the club newsletter SCALL. It should be noted however, that the newsletter is a club effort and may be discontinued at any time if support by the members drops below the current level.

For more information, contact the club secretary/treasurer, Thomas Scales, 5423 Dove Forest Ln, Humble, TX 77338. I have noted that SCALL carries good articles of interest, and may be worth considering subscribing to get that extra info about your H8 or H89 system.

WILLIAM FREY, 6106 Marlora Rd, Baltimore, MD 21239 (301) 323-6093 wishes to announce a local user group. William is president of the group, of about 45 members, operating out of the Heath Store in Towson MD every third Thursday. Anyone interested in joining this group, or wishing additional information, contact William.

James Czebiniak, 199 Vly Rd. Ext, Schenectady, NY 12309 is also interested in forming a development group (H8SCOOP FEB '81 REQUESTS). He says "Maybe we need a hotline network you can call when you have a problem. Names of people with expertise in certain areas. I feel I may qualify for giving advice in file and screen handling in MBASIC. My BASGEN program and documentation do a fairly thorough job of explaining how field statements work. I am willing to take calls and help people any way I can". Jim's phone number is (518) 869-8723.

Fred Pospeschill, 3108 Jackson St., Bellevue, NE 68005, (402) 291-0795 (7-9PM) writes "A Note to Other HA-8-3 Graphic Board Owners:

"Due to the rather limited initial production run there are only a few of us with this product - I seem to be the only one in the Omaha-Bellevue NE area. Although the ASM code delivered with the board is quite good, there are virtually no application programs available. So that we all don't have to invent the same wheels and solve the same problems, I thought that a graphics club/group which shared ideas, solutions, software, etc., might be beneficial. I would like to hear from others who may share similar thoughts. I am using TINY PASCAL at the moment and have Lucidata Pascal. Just to get things started, if you would like a copy of the routines I have developed thus far, send a disk and a dollar to cover postage. If you already have anything to share send it along on the disk. Using a clasp envelope which can be resealed will help keep the costs of sharing our efforts to an absolute minimum."

For any interested, Fred also purchased joy sticks from Jameco Electronics for \$5.95 each and has the info for a simple re-wire job to allow them to be used with the color board. This is the JVC-40 Joy Stick. For info on that, contact Fred.

MIKE PASQUALE, the one who presented the super device driver for the LA-36 device driver with 600 baud mod, is working on a hard disk drive for the H8. If anyone else is working in this area, you may want to keep in touch with Mike. Mike is also interested in double density and/or double sided 5 1/4 drives for the H8. Anyone working on this area, contact Mike at Yutana Barge Lines, 9750-3rd N.E. Suite 403, Seattle, WA 98155.

Bob McClure, 7521 Glenhill Rd, Austin, TX 78752 has the COOSOL 80 column printer. He paid \$495 and has it interfaced as AT: via H8-2 parallel board. It prints at about 40 lines per minute, which isn't bad. It can communicate with TTL RS 232 up to 9600 baud, and up to 6 print widths software selectable. It is capable of bi-directional printing. It appears to be a premium printer for it's price. If you have one and need info, or are planing to get one, Bob is the person to get in touch with.

Jeff Sohn, %JSM Communications, POB 118, Sheboygan Falls, WI 53085 has an EPSON MX-80 printer connected to his H89. This is a inexpensive compact dot matrix printer with many features. If you have one of these or want info, contact Jeff. He would like to hear from any others that have this printer.

REQUESTS

Robert Savage, captain in USAF is stationed at Robins AFB and wants assistance in locating H8 users in the middle Georgia Area. Write Robert Savage, 104 Van Dorn Ct, Warner Robins, GA 31093, or call him at (912) 922-6811.

Gordon Richards, POB 7211A, Orlando, FL 32854 has the following request: He is using the H8 to develop a computer controlled lawnmower. The mower uses a 8048 CPU and the H8 is used to make program mod's to the mower program developed on an INTEL development system. Gordon would like to find out if any subscribers to H8SCOOP have an 8048 assembler that will run on the H8, as this would make the project run smoother. Any help in this direction, please contact Gordon.

James Czebiniak (see WHO'S WHO for address) writes "We need some information on the internals of MBASIC; How arrays are stored and where, techniques to make programs more efficient and smaller, Do's and Don'ts, etc." Jim this may be coming through H8SCOOP. I know a guy who has disassembled MBASIC, and is presently writing a limited MBASIC compiler. Keep in touch with PROGRAMMING GOODIES.

Hugh Gregg, 2215 Empire Graded Rd, Santa Cruz, CA 95060, wants to know if anyone has any info on the availability of the Selectric Type ball with ASCII or quasi ASCII character set, particularly with [[] { < > and a few other odd such symbols all on one ball, and upper and lower case letters also.

INTERFACING WITH THE REAL WORLD A PARALLEL PORT TUTORIAL

Many reader questionnaires were returned with the desire for information of interfacing their computer with the outside world. Many individuals also have written in asking for information on this subject, so here it is.

I will present this as a three part article because of space limitations of this newsletter, the first part to cover GENERAL stuff, the second, INPUT to the computer and the third part to cover OUTPUT from the computer.

Since the most standard parallel port thus far available is the Heath H8-2 3 port parallel board for the H8, I will use this in my examples. Any type of 8 bit input and output port will be applicable, the main differences being the UART programming.

So what's a port? Think of a port as a doorway to the computer, allowing you to get stuff in or out. The H8 has 256 possible ports, since an 8 bit word

defines a port and $2^{**}8$ is 256. This can be contrasted to memory locations where a 16 bit word defines a unique address for a maximum of $2^{**}16$ or 65535 possible locations, commonly referred to as 64K.

What's the difference between a memory location and a port? Nothing really, except the memory location is addressed by a signal from the computer defined as MR for memory read (pin 28 on the H8 buss) and MW for memory write (pin 23 on the H8 buss), while a port is addressed by I/O R for Input/Output read (pin 26) and I/O W for Input/Output write (pin 21). Other than that, it's basically the same thing, just the way the computer tells them apart.

Ever hear of the term "memory mapped"? This is usually a clever way to use computer memory and its associated commands to access ports. For example, the 3K unused memory in the H8 from 004000 to 024000 could be used to input and output from the computer. These address locations would simply have to be "decoded", so when a memory read or write to them would occur, they would actually go somewhere besides memory. The dumb computer doesn't know, or care, whether there is memory sitting out there, or something else, or nothing!

An 8 bit word (Input or Output) from a port can be used directly to input or output eight different status points or commands directly. These could be decoded further, with up to 256 possible points for one eight bit word. A good example of this is my 16/32 channel ON/OFF controller where one 8 bit output port can controll up to 32 devices. This, however, is beyond the scope of this article. We will be dealing here with using the bits directly; how to input and output to them, and use them directly. The H8-2 3 port parallel I/O card by Heath will give you 8 input bits and 8 output bits per port with handshaking, for a total number of 24 inputs and 24 outputs. We will not go into handshaking here, because it will not be needed.

Just know that handshaking is simply communication between computer and associated connected devices to signal certain conditions; such as "I have data for you", "I am ready for data", "I have received the data", etc.

The Heath manual for the H8-2 parallel port is very scant in information. I spent many long nights when I got the board and started working on my selectric interface to figure it out. I finally called the Heath engineers, only to find out they were as much in the dark as I was, which is what I suspected from the start, thus the poor manual.

The H8-2 card contains a USART (universal synchronous-asynchronous receiver/transmitter) the 8251 for data conversion. Most interface cards contain this or a similar type of device (called an ACE for the H8-4 four port serial I/O card). I won't go into detail here why, or what it does, but it must be initialized before the card will work for port Input/Output. Don Long's Clock/Parallel I/O board has an onboard real time clock with battery backup plus 12 parallel ports, using the 8255 PPI chip (programmable peripheral interface). This is a different setup, but must also be initialized before it can be used.

Going back to the H8-2, each port channel occupies two consecutive computer ports, one for the data I/O (always the lower even number port) and the other for the USART initialization (the higher odd number port).

Next month I will first discuss how to initialize this card, then we will discuss the hardware and software involved for INPUT to the computer and what you can do with it.

* PROGRAMMING GOODIES *

A goodie sent in by Rick Comito. "I write my programs in modular fashion, and some of these modules make good utilities for use in other programs. To save just the module from the middle of a program, try this:

```
While in the command mode of BASIC--
type- OPEN "DEV:FNAME.MOD" FOR WRITE AS
      FILE #1
      LIST #1,linnum,linnum
      CLOSE #1
```

If you're careful about how you number your utility modules, you can include them easily in your BASIC programs from you disk of utilities while in HDOS by using the COPY feature:

```
COPY dev:newpgm.BAS=dev:oldpgm.BAS,dev:fname.MOD
```

The whole routine works best if you build your utilities with this use in mind, and keep similar utilities in fixed line number ranges. NOTE: If you specify LP: as dev:, it will give you a partial listing in hardcopy. A great debugging tool if you need to see more than 24 lines at a time. Don't forget to CLOSE the file, because you don't empty your buffer until you do.

How about this one for rounding off money. To keep money with no more than 2 decimal places to the right (cents), try this. Say you have \$17.95 with a sales tax of 6%. This tax would be \$1.077. What to do?

- 1) Multiply value by 100 to get rid of the two right decimal places.
 $1.077 * 100 = 107.7$
- 2) Add .5 to round up
 $107.7 + .5 = 108.2$
- 3) Take INT(n)
 $INT(108.2) = 108$
- 4) Divide by 100 to get decimal back
 $108 / 100 = 1.08$

EASY!

TECHNICAL FORUM

HDOS 2.0--HERE WE GO AGAIN

Well, now that we have almost totally automated HDOS 1.6, and have it working much the way we want, along comes 2.0. Naturally, most of the patch locations of 1.6 will not apply to 2.0, and many programs such as UNLOCK, FLAGS, SUBMIT and more

will not run under 2.0. It will take awhile to get the new data on 2.0 for patches etc., but when they are available, you will see them here.

My greatest dissapointment of 2.0 was the STATUS function; it no longer gives disk read/write errors. I considered this to be of the utmost importance, since by observing the number of soft errors since last boot you could pretty well determine how the Disk system was functioning. If the soft error rate began to climb, you automatically knew it was time to check disk rotational speed and step times. Now how will you ever know? (See OFFERS)

Jim Teixeira called and told me when you do disk reads, every time a soft error is found, a "!" is printed on the screen if you are using the H8-5 card. It turns out that this is only sent to the H8-5 (port 372Q) and is apparently a debug aid used by Heath during SY.DVD development. They forgot to delete the code from the distribution copy. (See the device driver code -SYDVD.ASM- at the label DSKERR). Anyway, if you have an H8-4 board and want to see the slashes, change location B1 in the same sector as above to E8 HEX (350Q). This change can be done at the source code level and re-assembly.

Another one sent in by Jim is the disk spin clock for version 2.0. It's now done in SY.DVD device driver rather than in HDOS as before. The timeout constant is located in the third sector, byte 55 HEX of SY.DVD. Use a DUMP program to locate the third sector (via DUMP SY.DVD). On Jim's distribution disk it was track 15 sector 2. Then use DUMP without any file name to change byte 55 (which is normally a 3C HEX for 30 seconds) to whatever value you want. Jim set his to 14 HEX for 10 seconds.

I have had problems with memory size. Programs that would run under 1.6 will not run on 2.0 because of the extra space the operating system takes up. This is only true of large programs that went to the memory limit under 1.6. I also noticed when I run the old MBASIC under 2.0, it prints some garbage on sign on. It has been reported by Al Noe that a small bug exists in 2.0 with SYSGEN. Al says it refused to SYSGEN properly using one drive, but worked OK if SY1: was designated for the destination drive.

ON NEW MBASIC

This version is larger than the previous one, and is reported to have several bugs in it. Some report bugs, some report no bugs. It is reported that it does run faster though. It may depend on the version of HDOS you use it with as far as the bugs go.

Al Noe writes in "I haven't played around with the new MBASIC yet, but see on Page 0-3 that the LP driver must still be loaded before calling MBASIC. According to page 2-58 of the new HDOS manual, COPY and CAT are the only valid LP: commands. An attempt to load LP: got me an immediate error message, and a quick run at MBASIC netted a disk I/O error on the first attempt to open LP.

ON DISK DRIVE PROBLEMS

A possible cure for disk drive problems was found in San Diego Heath User's Group newsletter, 12202 Kingsford Ct, El Cajon, CA 92021. "The Disk Drive would not read disk that had been DUPLICATED on another drive. I ran the General Drive Test and fainted. The results were 400 + soft errors on each pass and 1 to 3 hard errors. The solution was to apply two (2) drops of 3 in 1 oil on the stepper drive motor bearings which are located on the rear end of the screw drive shaft. The General Drive Test ran through without a single error after the oil was added. If you have had this problem just be careful not to over lubricate the Disk Drive."

From JERI-HUG newsletter comes some info about using the H-77 with the plastic insert in place of a drive for storing diskettes. "One of our members who has an H-77 with a single drive was using the plastic insert box mounted in the unit to store disks--Power line problems were experienced during a recent storm and the accompanying magnetic collapse caused the disks to be altered to the extent that they became unreadable. Luckily this member has backup for most of his data." Keep in mind, it is NEVER a good practice to store disks near any type of magnetic field. You're just asking for trouble-- Henry.

COMMENT ON H19 DISK DRIVE

EARL ALBIN writes in to tell me he has installed a disk drive in his H19 with no problems (see February H8SCOOP). He tells me "I got the H89 manual from Heath, and got all the shielding and mounting hardware that they use in the H89, and got a new Siemens drive from Heath also (I don't know why he didn't save over \$100 and go with Floppy Disk Services). I had to make up the longer ribbon cable, with the three needed edgecard sockets on it. I have a small muffin fan in the wooden cabinet the system is in to cool the heat sink on the rear of the H17, as this was running pretty warm before." So there you have it. There may be problems with shielding, but if you're careful, you should have no problems." Keep that in mind, H8ers, the H17 dual drive will work with the H89 or H88 (with disk controller card).

ON UP/DOWN LOADING

Quite a few individuals have written in requesting information on how to upload or download information from one computer to another, for example a Radio Shack to a H8 or H89. Presented here is the method for doing this.

DOWNLOADING is simply taking larger computer system programs and getting them in to a smaller computer. UPLOADING is going the other way, from a smaller system to a larger system. The procedure is the same. Note that getting the information in is one thing, but running the software is another. For example, the Radio Shack BASIC is very close to the HEATH MBASIC, but there are some differences. These will have to be edited or changed, before a program will successfully run. Also the TRS-80 handles graphics totally different than the H19/H89, so this would all have to be converted. There is no way to get the software in, press go, and make it

work. This is a system and software difference which must be modified for successful run.

The simplest way to get any computer's info into another is to get a listing and type it all in by hand. This is time consuming. Tying two computers together electronically will transfer ASCII files via serial ports, and get the information in a file for you, the speed dependent on the BAUD rate transfer you are using. The easiest way to do this is to have a MODEM at each end, using phone lines to exchange info. If you have two computers sitting side by side, you won't have to buy two modems, just hardwire the RS-232 ports together as described here. This is called a MODEM eliminator, or a NULL MODEM. Major contributions for this article were made available to me by Major D. C. Shoemaker.

Note that the function of the MODEM is to convert the RS-232 data signal obtained from the serial output port (which is quite standardized among computers) into something the phone line can pass. It converts the data into audio tones which the MODEM on the other end understands and converts back, which is understood by all other RS-232 equipment.

Before you rush off to try it, remember that there can be up to 30 volts at the port, and it's all too easy to wind up with this voltage in the wrong place, enough to blow one or more IC's. In addition, we have the question of half-duplex and full duplex, and the two need to be handled differently. The procedures are quite straightforward and pose no problem, but they must be followed to avoid embarrassing accidents. There are two figures that you should refer to as we proceed.

To determine the correct figure, you'll need to consider the MODEM driver you'll be using. Most require that you operate in either half-duplex or full-duplex, and in most cases there is no choice but to use what the program author calls for. Check documentation and be sure that all computers that you'll want to connect use the same mode!

For full-duplex (figure 1) the connections are very simple. Three wires are all that connect the two systems, with jumpers to connect the other pins at each end. The pin numbers are keyed to the DB-25 connectors usually associated with RS-232, but if your system is non-standard you can just refer to the description to make the appropriate connections.

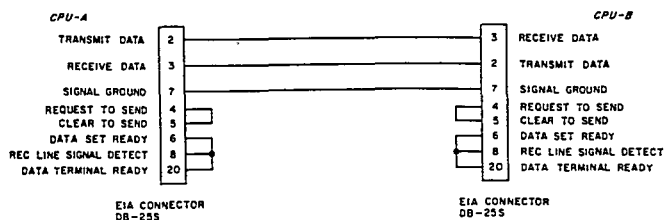


Fig. 1. Full-duplex modem eliminator.

If you use the half-duplex mode (figure 2) you'll have a bit more work. You'll need to insert small diodes as shown to insure that the desired signals go in the proper direction only. As shown, you'll need only two interconnecting wires, and as many

different computers can be connected as desired, up to a practical limit of four or five. Again, jumper the unused connectors.

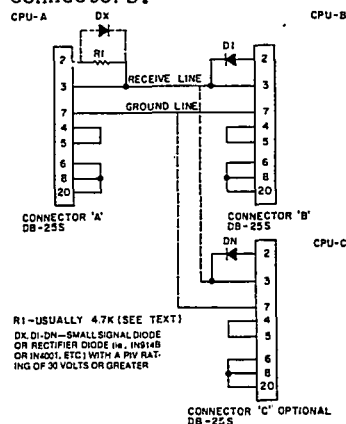


Fig. 2. Half-duplex modem eliminator.

Using either mode, this scheme allows up to five computers to share the same data. It makes no difference which one originates the data; the others will all receive the same thing. For instance, one operator selects a source code file that he wants to send out, and uses the resident MODEM program to open that file for transmission. Before pressing the final key to begin transmission the other operators use their own programs to open a file or establish a buffer in memory to receive the data. When they are all ready, the sender directs the data out. The others will all receive it at the same time, at the common baud rate (i.e. 300 BAUD) that they would operate at over the phone. When the transmission is finished, everyone closes files and moves on to the next program. The main feature of this approach is that several different systems can share the same files in short order, with the sole disadvantage they must all be in the same location.

A word on software. You will need a software package just like with a phone MODEM to handle the "housekeeping" and open and close files etc. Probably the best one to use for this is MCS, Available from HUG; 885-1050. The software sets BAUD rates, handles half or full duplex, establishes a buffer for input file, and takes care of all the goodies. Following is a sample session using MCS to allow a Heath H8 to communicate with a TRS-80, first sending a file to the TRS, then receiving one. Please note anything appearing within [], are for comments only.

>MCS [from hdos, run the MCS program]

MCS 01.03.04 [log on message]
ENTER "HELP" FOR ASSISTANCE

COMMAND=CONVERSATION [conversation mode]

.OK [ready prompt]

<CTRL-B> [exit to command model]
COMAND=SEND [send file out]

.OK

ENTER FILE NAME:LIFE.ABS [identify file]
TRANSMITTING.....
FILE HAS BEEN TRANSMITTED

COMMAND=OPEN [open h8 file]
 .OK
 ENTER FILE NAME:ACCOUNTS.BAS
 [name of h8 file]
 COMMAND=COPY [receive file]
 .OK
 DO YOU WANT A NEW BUFFER? YES
 [don't add to old buffer]

ENTER FUNCTION: [CR for TRS-80]
 COMMAND= [after transfer]

Walt Bilofsky of Software Toolworks informs me that his H89 REACH will nicely handle this task, at rates far above 300 baud, for the H89.

BRIEF BOOK REVIEW I

H-8 PROGRAMMING FOR BEGINNERS From dilithium press

You will be greatly misled if you read the book back cover "... it not only covers assembly language programming but also Benton Harbor Basic." Never is BH Basic mentioned, and although you conceive programs using mnemonics that are introduced and explained, the book covers machine language, and NOT assembly language!

Other than that, I'd recommend the book to anyone with an H8 who wants to learn the actual machine language of the H8 8080 CPU. The book deals with the H8 and communicating with the processor via the front panel keyboard and display in OCTAL, which is what the H8 was designed for anyway, not HEX as is creeping up all over the place today (for some strange reason). You do as you go, entering programs via front panel and observing results; more like a programmed learning approach.

BRIEF BOOK REVIEW II

INSTANT BASIC by Jerald R. Brown (dilithium press)

INSTANT BASIC is a book covering most beginning aspects of a general type of Microsoft Basic. There is a lot of space taken up with pictures, art, "funny stuff?", and other irrelevant material to BASIC programming, perhaps to attempt to make the book more interesting or cute?

For the beginner knowing little or nothing about Basic, I would say this book will give a good start to understanding the more common standard MBASIC commands, and would apply about 90% to BH BASIC.

For the experienced programmer in BASIC or MBASIC, you will gain little or nothing from this book. Ken Knecht's MICROSOFT BASIC is still tops on my list.

"The author has a quiet writing style that slowly introduces new ideas in a pleasant non-mathematical context. To offset this style of writing, he has used the zaniest, wildest, graphics available making the book a barrel of fun to read!" (That's debatable--Henry).

GOOD GUYS and BAD GUYS

BAD GUYS--ADVANCED COMPUTER PRODUCTS, POB 1739 Irvine, CA 92713. They were also listed in July 1980 H8SCOOP for bad guys, so I guess there's a lot of trouble with them.

John writes "I ordered a single part (AY-3-8910) on August 15th. I did not receive the part or any acknowledgement of my order, but my check was cashed. About a month after the order, I sent a letter asking that my order be investigated. I received no response. On October 6th, I called on the order line, but was told I would have to call another (non toll free) number, so I wrote again asking whether the company might be another World Power Systems and enclosed a copy of the letter to the Wisconsin Attorney General's office. This got the desired reply as they sent the part (without any explanation) about a week later."

Let's try to avoid places with this type of customer service, they don't deserve our business. There's plenty of GOOD places out there that go out of their way to help, and ACP is not one of them!

GOOD GUYS--Greg Saville who offers the PSG sound board (See August 1980 H8SCOOP) and CHEAPTALK (See December 1980 H8SCOOP) for the H8. It is reported that not only does Greg have prompt service, but he stands behind his products. Keep it up Greg, we need more like you.

GOOD GUYS--Walt Bilofsky of THE SOFTWARE TOOLWORKS. Walt has been given this award before (Aug 1980 H8SCOOP), but I continue to get letters about his high quality software and exceptional service.

BAD GUYS--CYBERSOFT, 6400 Gila Ct, Plano TX 75023. James Czebiniak writes in "I bought a \$25 game disk only to find that the programs were crap. There were 5 games with no graphics and they were on the same level as the first release of HUG Volume 1. I was a little peeved so I wrote them and received no reply."

BAD GUYS--THE CPU SHOP, Charlestown, MA. Robert McClure writes in "I ordered a box of ten Verbatim disks from the CPU shop from their ad in Jan '81 Microcomputing on January 5. I used the toll-free number and master charge. They said allow 14 days to ship. Right there I should have cancelled the order. I had just bought a used H17 Dual Disk and knew it would be shipped shortly, so I figured the H17 might beat the disks to my door, but not by much.

"Here it is, January 22nd and I have an H17 but no blank disks. I called the CPU shop and found they had been out of ten-sector disks and just got some shipped them "a couple days ago". How long to wait now? "Give it another week". Seems they shipped one box by UPS. Why not first-class mail? I would think my \$3.00 shipping charges would cover that. "We just deal with one shipper."

"Point 1: Why didn't they tell me they were out of ten-sector disks? Then I would not have ordered from them. Point 2: I don't see why I should not be able to get the disks in a week by first-class mail for the same (or less) shipping charges. Bottom Line: The disks are not \$24 for 10. They are \$27 for 10 including slow shipping. CAVEAT EMPTOR."

GOOD GUYS--ROBERT PEARCE, 504 McCoy's Fork Fd. Walton, KY 41094 (606) 485-4951. "Robert has done some nice things to SCLEBI's PERSONAL INFORMATION MANAGEMENT SYSTEM (PIMS). Upon contacting him he sent me a copy of his improved program along with a couple others. He also sent along a listing, as I don't yet have a printer. He did this for only a blank disk. I had used PIMS on a PET and was attempting to work it into my H8/H17/H19 when I learned of his effort. Boy am I glad I got in touch with him. He saved me lots of time and showed me a couple of things I probably never would have thought of on my own." So there you have it. This sent in by P. Pitzer. If any of you are into this, or want to know more about it, write Robert Pearce.

CLASSIFIEDS

FOR SALE--HA-8-2 Music Synthesizer system for H8. Complete with board, manual, software and additional music. \$110 prepaid UPS. Allan Moser, Rt. 1, Box 16, Samuels, ID 83862. (208) 263-3726.

AT PRESS TIME

Pardon the format of this newsletter, I wanted to make it extra big to take full advantage of the extra postage I must pay on this issue with the larger size and new postal rates (bless their little hearts). What would have cost 28 cents to send out, is now costing me 35 cents, an increase of 8 cents per issue. I therefore reasoned with myself to bring this issue as close to the 2 ounce limit as I can, to get my (and your) money's worth. I had the whole newsletter put together and formatted, and this is being added as an after thought, so take it for what it's worth. You're getting this extra stuff that would not normally have gone out till next month.

DIABLO FEATURES VIA SOFTWARE CONTROL

Since there are many Diablo owners out there, and seems to be more all the time, I wanted to present some good stuff for you to use, to control some of the Diablo's excellent features. I was under the impression you had to have the word processing option installed to make use of BOLD and SHADOW print, and other features. Not true! Craig Todtschinder of THE SOFTWARE TRADER sent me a program to demonstrate some of the features that can be software controlled. I modified this program and added BOLD type printing and here's the final result. It is written for BH EXT BASIC, but can easily be modified by changing OPEN FILE line (110, 320) to run under MBASIC.

WANTED--A good used WANGO Model 82 5 1/4" drive. Will consider an entire H17 with WANGO drives if the price is right. I don't need the controller card. John Doak, 45 Allen Dr., Woodstock, NY 12498. (914) 679-8723.

WANTED--2 Heath 8K RAM boards. Will pay up to \$70 each. Contact Steve Evans, 105 Altadena Dr, Mt. Lebanon, PA 15228. (412) 344-5264.

FOR SALE--ADDS REGENT 100 (with options). CRT in excellent condition. \$795. Two TELETYPE Model 33 for \$200 each. Tom Roche, 29W476 Blackthorn Ln, Warrenville, IL 60555. (312) 393-2023.

FOR SALE 2 H8-2 3 port parallel ports for \$125 each. In excellent condition, tested and burned in. Contact H8SCOOP.

FOR SALE 2 8K memory boards for the H8 computer at \$60 each, you pay shipping. Contact John Rossmann, 9804 W. Morgan Ave., Milwaukee WI 53228. (414) 327-1772.

FOR SALE OR TRADE--A 16K Heath WH-8-16 Memory board at \$150 or will trade for a Heath H-17-1 disk drive. Less than 50 hours on the board. Contact Ted Elzerman, 3 Settler's Ln, Springfield, IL 62707. Call (217) 546-8598 after 6PM CST.

```
00010 REM ** DIABLO FEATURE DEMO BASIC PROGRAM
00015 REM ** BY CRAIG TODTSCHINDER AND HENRY
00017 REM ** FALE FOR H8SCOOP
00030 PRINT "(1)..SET PITCH TO 10"
00040 PRINT "(2)..RETURN CONTROL OF PITCH SWITCH"
00050 PRINT "(3)..SET LINES PER INCH TO 6"
00060 PRINT "(4)..SET LINES PER INCH TO 8"
00070 PRINT "(5)..PRINT A LINE--SHADOW PRINTED"
00080 PRINT "(6)..RETURN TO OPERATING SYSTEM"
00090 INPUT "SELECT ";A
00095 IF A<6 THEN OPEN "LP:" FOR WRITE AS FILE #1
00100 ON A GOTO 120,170,220,280,330,500
00120 PRINT #1,CHR$(27);CHR$(31);CHR$(13);
00125 REM ** ESC;US;CR TO SET PITCH TO 10
00130 GOTO 430
00170 PRINT #1,CHR$(27);CHR$(83);
00175 REM ** RETURN TO NORMAL SETTING--ESC;S
00180 GOTO 430
00220 PRINT #1,CHR$(27);CHR$(30);CHR$(7);
00225 REM ** ESC;RS;BEL--SET 6 LPI
00230 REM ** THE LAST CHARACTER (BEL) IS VARIED
      TO CHANGE LINE SPACING.
00240 GOTO 430
00280 PRINT #1,CHR$(27);CHR$(30);CHR$(8);
00285 REM ** ESC;RS;BS;--SET 8 LPI
00290 GOTO 430
00330 PRINT "ENTER LINE TO BE SHADOW PRINTED"
00340 LINE INPUT ;R$
00350 REM ** SHADOW PRINTING RECOMENDED AT 10
      CHARACTERS PER INCH
00360 PRINT #1, R$;:REM** PRINT LINE ONCE
00370 PRINT #1,CHR$(13);:REM** RETURN, NO LF
00371 REM ** THE NUMBER N CAN BE VARIED FOR
      DIFFERENT SHADOWS FROM 1 FOR REGULAR,
00373 REM ** 2- FOR BOLD, 4 OR 5 FOR DEEP SHADOWS.
```

```

00379 INPUT "ENTER SHADOW DESIRED, 2 FOR LITTLE 5
      FOR MUCH ";N
00380 PRINT #1,CHR$(27);CHR$(31);CHR$(N);" ";
00385 REM ** ESC;US;OFFSET;SPACE
00390 PRINT #1,CHR$(27);CHR$(83);
00495 REM ** RETURN TO NORMAL SPACING
00400 PRINT #1,R$:REM ** REPRINT LINE WITH OFFSET
00410 CLOSE #1:GOTO 30
00430 PRINT #1,"THIS IS A TEST"
00440 PRINT #1,"THIS IS A TEST"
00450 PRINT #1,"THIS IS A TEST":CLOSE #1:GOTO 30
00500 END

```

These software commands open new doors for the Diablo. BOLD PRINT is nice for making headings or labels, and SHADOW PRINTING has it's good results also, when used properly. This is just the start, Vertical movement can be likewise controlled, and so can many other features. Using this as a guide, you should be able to discover more for yourself.

REQUEST FOR HELP IN ILLINOIS

Ralph Munroe Jr, 359 Miami St, Park Forest, IL 60466 (312) 748-2171 wants to know if there is anyone in his area with an H8 or H89 setup that he could depend on for equipment use or rental for short periods of time should his system ever die. Ralph runs business applications, and is concerned about a backup system. If anyone in this area is interested in exchanging information, contact Ralph. I'm sure a mutual agreement could be worked out. Ralph runs an H8 with 56K, dual H17, H14 and H19.

ON DISKS AND DRIVES

Dan Rodenz, 66 13th Ave. SW, New Brighton, St. Paul, MN 55112 brings up an interesting question on 4 5 1/4" drives in HDOS. Dan says a 2nd older H17 dual drive would solve housing and power supply problems very nicely and give more storage.

"The Wanco operators manual for my H17 drives clearly state that these drives can be chained together in 2, 3, or 4 drive configurations; it is easy to address up to 4 different drives with the selection chip on the PC boards. Presumably the controller board doesn't care whether 1, 2, 3, or 4 drives are connected to it. The only problem then, as I see it, is HDOS software."

Anyone knowing about this one, contact Dan. I feel by going to 4-5 1/4" drives, one could just about purchase a single 8" drive giving 1 Megabyte storage for about the same price, and have over double the storage.

Walt Bilofsky writes in some interesting points on Hard disk vs Floppy disks, some points very well taken. "Here's my comment on 8" vs. hard disk drives (with thanks to Jim Gillogly for raising some of these points): I agree with Henry Fale that hard drives are or will shortly be priced competitive with 8" drives, especially Heath's. But, 8" floppies give you the following four advantages over hard disks.

"First, they solve the backup and archiving problem with no additional hardware. How do you back up a 5 megabyte disk? Second, with 8" single density CP/M disks you can exchange data with and CP/M user, Heath or non-Heath, who has 8" disks. Hard to mail a Winchester platter.

"Third, how do you organize a hard disk? Imagine the joy of doing a CAT or DIR and seeing four hundred file names. You need some kind of compartmented directory, which HDOS lacks, and CP/M provides (with user letters) but could do better. With Floppies, you simply use one disk per project.

"Fourth, how many of us really need more than a couple of megabytes on line at once? And for those that do, how long is 5 megabytes going to suffice? When it comes time to expand, it's easier to buy a box of floppy disks than another hard disk drive. Floppies provide an infinitely expandable data bank, not "on-line" to be sure, but no more than a disk change away.

"Hard disks are nice if you really need super fast data access, and indispensable for multiple user timesharing systems. They're necessary if your application contains a multi-megabyte database. But for most uses of H8 and H89 machines, I think 8" double sided double density floppies are preferable."

Again, as I said, some points well taken, I bite my tongue!

FLIPPY FLOPPY?

I have been asked to comment on flipping the 5 1/4" disks over in the drive to use the second side (after modifying the disk or using a flippy drive of course). I have been negative about this in the past, and continue to be.

I print something sent in to me by Dave Heine. "At a recent meeting of the San Jose HUG, I asked if anyone had any experience with punching out new holes in the disks and using the reverse side. This generated over half-hour's worth of discussion.

"Some said they had been doing it with no problems; others denounced it as the "worse" thing you could do! Several people where I work have been doing this for several years with their S-100 based systems with no problems. I have compromised by doing it only with the "library" copies of disks I have that are seldom used. So far I have converted over 20 disks and have yet to find a bad sector on the reverse side. I am using BASF and MEMOREX disks. (In fact, the bad sectors I have found have shown up on the normal sides!) I do not use them for the frequently used system and scratch disks. I do not use any special tools other than a standard hold punch and an xacto knife. The hole punch I have has a small plastic cover that catches the hole after it is punched out so that it doesn't drop down inside the disk cover. I use another disk

as a template to mark the position of the hole and the notch on the side. After punching the necessary holes I always run a Media Test on them (both sides) to be sure they are OK.

"One last thought--if dust, etc. is such a problem with disks why do many manufactures have a cooling fan blowing air (and dust particles) through the drive mechanism? Experience shows that if one exercises reasonable care he should not encounter disk going bad very often. Important, irreplaceable data on disks should be backed up anyway."

Now my comments. Any good disk manufacturer that has a fan in the installation also has filters to catch the dust from the air. A few years back when disks cost over \$5 each, I would say maybe, but now since they are around \$2 to \$3, it seems to me an awful gamble to take. First off, the ordinary disks are not certified for double sided operation, if they were, the holes would already be punched and you'd pay more for them.

Second, the disk will rotate in an opposite direction when inserted on the other side, which may cause trapped dirt, dust and oxide particles to come loose from the sleeve jacket, which when turned again, may clog the read/write head. While this may not show up at the time, it may when you are doing an important data dump on a disk, only to find you have lost it all! Then where is the backup? It doesn't exist yet!

I suppose as Dan says, if each disk is tested before use by running a Media test on it, and back-up's are kept of everything, it may be acceptable for some. As I stated before, for the cheap price of disks, I would rather not take the gamble. Anymore comments on this, direct them to H8SCOOP.

Henry

ON MODEM ARTICLE

For additional information on null modems, see Byte, February 1981 page 198, "BUILD A NULL MODEM". Also refer to Microcomputing, February 1981 page 112.

SPACE FILLER

As long as I finally have some space left, let me tell you what's coming. First off, I will continue the interfacing article next month. I will make every effort to have next month some information and programs on SORTING, since it is one of the computers major tasks. I will describe different sorting programs and computer languages, vs speed.

I'm almost finished with my report on Word Processors and Text Editors. This will be out next month if there's room otherwise the following month.

I have a project sitting here submitted by Wayne Frick on Computer Graphics for the H8 using an X-Y oscilloscope, a breadboard card, and about \$15 worth of parts. This will probably be printed in two or three parts, depending on the space situation.

I had a call from John Merrill tonight telling me the HA-8-3 0 org card for the H8 will NOT work with over 48K of memory, and a fix is underway with an expected shipping date in about 3 weeks. If anyone has info on this, PLEASE let me know at once, before I butcher my computer with mods only to find it will not work.

There has been lot of request for interfacing the H8 to various varieties of home video games. Anyone working in this area, contact H8SCOOP. the end

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